



Federal Office  
for Information Security

BSI Technical Guideline TR-03135-2

# Machine Authentication of MRTDs for Public Sector Applications

Part 2: Application profiles for official document inspection systems

Version 5.0.1



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# 1 Changelog Part 2

The following tables present the changes introduced between the latest versions of this Technical Guideline. The changelog lists the changes grouped per part of this Technical Guideline, per element (section, table, figure) and type of change (refer to [KeepAChangelog]):

- *Added* for new features
- *Changed* for changes in existing functionality
- *Deprecated* for soon-to-be removed features
- *Removed* for now removed features
- *Fixed* for any bug fixes
- *Security* in case of vulnerabilities

## 1.1 Version 5.0.1

| Element Name | Type of Change | Change Description |
|--------------|----------------|--------------------|
|--------------|----------------|--------------------|

**Table 1.1** Changelog Version 5.0.1

## 1.2 Version 5.0.0

| Element Name                                                    | Type of Change | Change Description                                                                                                                          |
|-----------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Chapter Operational Scenarios                                   | Changed        | Changed name to Application Scenarios                                                                                                       |
| Changed Application Scenarios and                               | Changed        | Switched to capability-based approach for the different scenarios.                                                                          |
| Inspection system capabilities - Inspection system capabilities | Changed        | Changed blacklist to whitelist in case of successful electronic checks in option to attenuate the influence of spectrally selective checks. |
| Chapter introduction                                            | Added          | Added section on publication language                                                                                                       |
| Chapter Requirements from TR-03135-1                            | Changed        | Added requirements on transactional logging                                                                                                 |

**Table 1.2** Version 5.0.0

## 1.3 Version 2.5

| Element Name                         | Type of Change | Change Description                                                    |
|--------------------------------------|----------------|-----------------------------------------------------------------------|
| Chapter Requirements from TR-03135-1 | Added          | Digital seal checks as general requirement                            |
| Chapter Requirements from TR-03135-1 | Added          | Digital seal checks requirements for all application scenarios        |
| Chapter Requirements from TR-03135-1 | Added          | Extended data nodes for digital seal checks                           |
| Chapter Requirements from TR-03135-1 | Added          | Standard logging profile with evaluation extensions for digital seals |
| Chapter Requirements from TR-03135-1 | Added          | Asylum registration application scenarios                             |

**Table 1.3** Version 2.5

## 2 Introduction

This document is a part of [BSI TR-03135]. [BSI TR-03135] specifies and describes necessary requirements for machine assisted document checks on Machine Readable Travel Documents (MRTDs) in Public Sector Applications.

### 2.1 How to use this document

1. Comply with the requirements for all document inspection systems in ▶4.1: Requirements for all inspection systems.
2. Identify your application scenario in ▶3.1: Application Scenarios Catalogue.
3. Comply with the requirements for your application scenario in ▶4.2: Requirements for Application Scenarios.

### 2.2 Terminology

#### 2.2.1 Key Words

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

#### 2.2.2 Definitions

##### 2.2.2.1 Capability

The ability to perform a specific task or action that is relevant for a document inspection process.

##### 2.2.2.2 Application Scenario

A set of circumstances that describes where, how, when and/or with what objective a document inspection system is used.

### 2.3 Abbreviations

For clarity and better reading, terms are abbreviated. This document contains the list of the used abbreviations at the end.

### 2.4 Publication Language

This Technical Guideline is only published in English. The reasons for this are as follows:

- Relation to other English only publications (esp. [ICAO9303], [BSI TR-03129-2], [BSI TR-03121] and [BSI TR-03110])
- The content is relevant in a European context.

## 3 Catalogues

This chapter lists all application scenarios, capabilities and checks which are relevant for this document.

### 3.1 Application Scenarios Catalogue

The following subsections describe all application scenarios, which occur in German Public Sector Applications.

#### 3.1.1 Application scenario - Stationary

Stationary systems are systems which are permanently installed at border crossings, at police stations or at other localities with raised focus concerning security issues (e. a. government buildings, ministries, embassies or official meeting locations) as well as temporary systems in accordingly prepared sites (e.g. for major events such as sports or music events).

#### 3.1.2 Application scenario - Self Service

Self Service systems are systems with which automated or semi-automated document and identity checks are performed (e. g. so-called eGates/Automated Border Control (ABC) systems, automated border control systems/gates).

#### 3.1.3 Application scenario - Partially mobile

Partially mobile application scenarios usually take place at checkpoints, in patrol cars or vessels, but also in rail, air and shipping traffic environments. This includes, for example portable PC stations, which are the size of a notebook, or a tablet PC. These systems are often equipped with a swipe or full-page reader in order to be able to read the Machine Readable Zone (MRZ)/Card Access Number (CAN), but also with a Radio-Frequency Identification (RFID)-chip reading module.

#### 3.1.4 Application scenario - Fully mobile

Fully mobile application scenarios will usually be performed in rail, air and shipping traffic as well as during general patrol duty. The systems are carried by the officer. The devices have the similar size and design as smartphones or tablet PCs.

#### 3.1.5 Application scenario - Kiosk

Kiosk application scenarios usually take place at border crossing points (air- or sea-ports). These systems are unengaged and unmanaged pillar styled units that can be operated by the traveller. A traveller can pre-register and pre-process before crossing the border. These systems are equipped with a full-page document reader in order to be able fully read the MRTD. This also includes reading the RFID chip of an Electronic Machine Readable Travel Document (eMRTD). In addition, such a kiosk system can capture live images of the traveller and fingerprints. An interactive electronic registration prompt guides the traveller through the process.

#### 3.1.6 Application scenario - Asylum registration

Asylum registration application scenarios take place when a human seeking asylum is registered.

### 3.2 Capabilities Catalogue

The following subsections describe all capabilities that are relevant in the context of [BSI TR-03135].

#### 3.2.1 Capability SCAN\_MRZ

The ability to detect the presence of a MRZ on a document and produce a digital representation of its content.

#### 3.2.2 Capability SCAN\_CAN

The ability to detect the presence of a CAN on a document and produce a digital representation of its content.

### 3.2.3 Capability SCAN\_DIGITAL\_SEAL

The ability to detect the presence of a digital seal on a document and produce a digital representation of its content.

### 3.2.4 Capability CAPTURE\_FULL\_PAGE\_VI

The ability to capture a full-page image of a document under white light (VI) lighting.

### 3.2.5 Capability CAPTURE\_FULL\_PAGE\_IR

The ability to capture a full-page image of a document under infrared (IR) lighting.

### 3.2.6 Capability CAPTURE\_FULL\_PAGE\_UV

The ability to capture a full-page image of a document under ultraviolet (UV) lighting.

### 3.2.7 Capability READ\_ELECTRONIC

The ability to read electronic documents without contact.

### 3.2.8 Capability COMBINE\_DOCUMENT\_COMPONENTS

The ability to combine information of different document components.

### 3.2.9 Capability COMPARE\_DOCUMENTS

The ability to compare two related documents.

## 3.3 Checks Catalogue

The following subsections describe all checks, which are relevant for German Public Sector Applications.

### 3.3.1 Check - Document

As specified in [BSI TR-03135-1],  Chapter 5: Authentication of MRTDs.

The implementation SHALL comply with the requirements listed in ►4.4.1: Requirements for Check - Document .

### 3.3.2 Check - Optical

As specified in [BSI TR-03135-1],  Section 5.7: Checking an optical component.

The implementation SHALL comply with the requirements listed in ►4.4.2: Requirements for Check - Optical .

### 3.3.3 Check - Document Detection

As specified in [BSI TR-03135-1],  Section 5.7.3.1: Document detection and capturing.

### 3.3.4 Check - Document Model Identification

As specified in [BSI TR-03135-1],  Section 5.7.3.2: Document model identification.

### 3.3.5 Check - MRZ consistency

As specified in [BSI TR-03135-1],  Section 5.7.3.9.1: Check the consistency of MRZ data.

The implementation SHALL comply with the requirements listed in ►4.4.9: Requirements for Check - MRZ consistency .

### 3.3.6 Check - MRZ IR readability

As specified in [BSI TR-03135-1],  Section 5.7.3.9.2: Check MRZ to be readable under IR light.



The implementation SHALL comply with the requirements listed in ►4.4.10: Requirements for Check - MRZ IR readability .

### 3.3.7 Check - MRZ OCRB type face

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.9.3: Check MRZ to be printed using OCR-B type face.

The implementation SHALL comply with the requirements listed in ►4.4.11: Requirements for Check - MRZ OCRB type face .

### 3.3.8 Check - MRZ alignment

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.9.4: Check alignment of MRZ.

The implementation SHALL comply with the requirements listed in ►4.4.12: Requirements for Check - MRZ alignment .

### 3.3.9 Check - Issuer versus Nationality

As specified in [BSI TR-03135-1], ☞ Section 5.6.1.2.7: Check the consistency between issuer and nationality.

The implementation SHALL comply with the requirements listed in ►4.4.4: Requirements for Check - Issuer versus Nationality .

### 3.3.10 Check - No Test Document

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.4: Check for test document.

The implementation SHALL comply with the requirements listed in ►4.4.5: Requirements for Check - No Test Document .

### 3.3.11 Check - Primary Facial Image Embedding

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.8: Check for the embedding of the primary facial image.

The implementation SHALL comply with the requirements listed in ►4.4.6: Requirements for Check - Primary Facial Image Embedding .

### 3.3.12 Check - MRZ versus Visual Zone

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.11: Check for the consistency of MRZ and VIZ.

The implementation SHALL comply with the requirements listed in ►4.4.7: Requirements for Check - MRZ versus Visual Zone .

### 3.3.13 Check - Visual Zone Consistency

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.10: VIZ consistency check.

The implementation SHALL comply with the requirements listed in ►4.4.8: Requirements for Check - Visual Zone Consistency .

### 3.3.14 Check - Spectrally Selective Verification

As specified in [BSI TR-03135-1], ☞ Section 5.7.3.12: Spectrally selective verification.

The implementation SHALL comply with the requirements listed in ►4.4.3: Requirements for Check - Spectrally Selective Verification .

### 3.3.15 Check - Electronic

As specified in [BSI TR-03135-1], ☞ Section 5.8: Checking an electronic component.

The implementation SHALL comply with the requirements listed in ►4.4.13: Requirements for Check - Electronic .

### 3.3.16 Check - Authenticity

As specified in [BSI TR-03135-1],  Section 5.8.5: Check the authenticity.

The implementation SHALL comply with the requirements listed in ►4.4.14: Requirements for Check - Authenticity .

### 3.3.17 Check - CSCA Validity

As specified in [BSI TR-03135-1],  Section 5.6.3.2.3: Check the country signer certificate's validity period .

The implementation SHALL comply with the requirements listed in ►4.4.15: Requirements for Check - CSCA Validity .

### 3.3.18 Check - Digital Seal

As specified in [BSI TR-03135-1],  Section 5.9: Checking a digital seal component.

The implementation SHALL comply with the requirements listed in ►4.4.16: Requirements for Check - Digital Seal .

### 3.3.19 Check - Combined

As specified in [BSI TR-03135-1],  Section 5.10: Combined checks.

### 3.3.20 Check - Cross Document

As specified in [BSI TR-03135-1],  Section 5.11: Cross document checks.

## 4 Requirements

This chapter details the requirements from [BSI TR-03135-1] which are relevant for German Public Sector Applications.

### 4.1 Requirements for all inspection systems

The generic requirements of an Inspection System (IS) SHALL fit the requirements regarding  Chapter 4: Inspection System Requirements and  Chapter 5: Authentication of MRTDs in [BSI TR-03135-1].

#### 4.1.1 Requirements on the execution of all checks

If any check is required to be performed in a specific application scenario and expected to be applicable to the identified document model, all partial checks of that check that are expected to be applicable to the identified document model SHALL be performed, unless stated otherwise. All checks SHALL be performed with the `runMode` set to mandatory, unless stated otherwise.

#### 4.1.2 Requirements for the retrieval of certificate lists

There are two scenarios for retrieving a new list of the type Master List or Signer List.

##### 4.1.2.1 Manual Retrieval and/or Deployment

If the list has to be retrieved and/or deployed manually, the administrator of the document inspection system SHALL retrieve and deploy the most recent list at least once every four calendar weeks.

##### 4.1.2.2 Automatic Retrieval and Deployment

The RECOMMENDED implementation is described in [BSI TR-03135-1] section 5.6.1.1.

If the list can be automatically retrieved and deployed by the document inspection system, it SHALL retrieve and deploy the most recent list at least once every three workdays. It is RECOMMENDED to check for updates daily and to deploy them immediately.

#### 4.1.3 Requirements on operational monitoring

The document check system SHALL implement the logging schema according to [BSI TR-03135-1],  Chapter 6: Logging scheme for operational monitoring.

This document defines the following levels of logging configurations:

1. Basic logging profile
2. Standard logging profile
3. Standard logging profile with optical evaluation extensions
4. Standard logging profile with electronic evaluation extensions
5. Standard logging profile with digital seal evaluation extensions
6. Full logging profile

The document check system SHALL implement all defined logging configurations. Setting the actual logging configuration SHALL be configurable by the application.

An extended data node is a data node that uses a type from the `bsi-tr-additionaldata` schema. The extended data nodes for a certain check are all extended data nodes that are part of the check.

##### 4.1.3.1 Basic logging profile

Basic logging SHALL contain all non-personally identifiable information from a check, including all check results. All data items from the Extensible Markup Language (XML) schema SHALL be filled, with the exception of the set of extended data nodes.

### 4.1.3.2 Standard logging profile

Standard logging SHALL contain all available information from Basic logging profile and SHALL provide additional information on the specific **failed** and **undetermined** check(s).

For electronic check errors, the optical document data, i.e. all extended data nodes from `dco:ProcessInformation` and `dco:ComponentInformation`, SHALL be included to allow for cross-comparison of optical and electronic document data.

Depending on the type of error, the relevant sets of extended data nodes detailing the error SHALL be present in the log. Note that extended data nodes are not present if the corresponding check was aborted. If the integrity check of a data group failed, then the extended data nodes of the optical check SHALL also be included.

### 4.1.3.3 Standard logging profile with optical evaluation extensions

Standard logging with optical evaluation extensions SHALL contain all information from Standard logging and SHALL contain all extended data nodes for the optical check regardless of the optical check result. Additionally, it SHALL contain all extended data nodes for the combined check and the cross document check that relate to the target of the evaluation.

### 4.1.3.4 Standard logging profile with electronic evaluation extensions

Standard logging with electronic evaluation extensions SHALL contain all information from Standard logging and SHALL contain all extended data nodes for the electronic check regardless of the electronic check result. Additionally, it SHALL contain all extended data nodes for the combined check and the cross document check that relate to the target of the evaluation.

### 4.1.3.5 Standard logging profile with digital seal evaluation extensions

Standard logging with digital seal evaluation extensions SHALL contain all information from Standard logging and SHALL contain all extended data nodes for the digital seal check regardless of the digital seal check result. Additionally, it SHALL contain all extended data nodes for the combined check and the cross document check that relate to the target of the evaluation.

### 4.1.3.6 Full logging profile

Full logging SHALL contain all available information including all extended nodes, regardless of the check results.

## 4.1.4 Requirements for data transmission

The IS SHALL implement the transmission and log specification according to [BSI TR-03135-1], Chapter 6: Logging scheme for operational monitoring for Application scenario - Stationary, Application scenario - Self Service, Application scenario - Kiosk and Application scenario - Asylum registration and it is RECOMMENDED for the ISTo to implement it for the Application scenario - Partially mobile and Application scenario - Fully mobile application scenarios. Logging data SHOULD be temporarily kept local, if transmission fails and retransmitted if the connection is available, possibility to manually transfer logs is RECOMMENDED.

## 4.1.5 Requirements on transactional logging

The logging schema defined in this Technical Guideline can be used for the sole purpose of document inspection (stand-alone variant) but also in the wider context of identity management (payload variant) which also consists of biometrics and process management, among other things.

For the stand-alone variant, the logging schema provides a transaction node below which all other checks are grouped. The transaction can be used to log messages in the scope of document inspection. In addition to checks of individual documents, checks between documents (cross-document checks) can also be logged.

One example of the payload variant is the logging in [BSI TR-03156]. When using this logging in identity management, no transaction node from the document check is used. Instead, the checks are transported via the so called `psidmlog`. In this case, the log is a container for the document inspection logs.

In the following, Table 4.1 shows where the document check and the cross-document checks are located in each of the two cases.

| Log                           | Check                | Container                 | XPath                                                       |
|-------------------------------|----------------------|---------------------------|-------------------------------------------------------------|
| BSI TR-03135<br>(stand-alone) | Document Check       | /ct:CheckTransaction      | /ct:CheckTransaction/ct:DocumentChecks                      |
|                               | Cross Document Check | /ct:CheckTransaction      | /ct:CheckTransaction/ct:CrossDocumentChecks                 |
| BSI TR-03156<br>(payload)     | Document Check       | /idobjlog:IdentityObjects | /idobjlog:IdentityObjects/idobjlog:DocumentCheckObject      |
|                               | Cross Document Check | /idobjlog:IdentityObjects | /idobjlog:IdentityObjects/idobjlog:CrossDocumentCheckObject |

**Table 4.1** Transactional logging

## 4.1.6 Requirements on time measurements

Time measurements SHALL be logged regardless of the cardinality used in the schema if the corresponding process was performed. Timestamps and durations SHALL use millisecond accuracy.

## 4.2 Requirements for Application Scenarios

The following sections describe which application scenarios require which capabilities.

If additional capabilities are present, the execution of the related checks (►Section 4.3) is RECOMMENDED, but SHALL be decided by the operator.

### 4.2.1 Requirements for Application scenario - Stationary ►3.1.1

The required capabilities for this scenario are:

- ►3.2.1: Capability SCAN\_MRZ
- ►3.2.2: Capability SCAN\_CAN
- ►3.2.3: Capability SCAN\_DIGITAL\_SEAL
- ►3.2.4: Capability CAPTURE\_FULL\_PAGE\_VI
- ►3.2.5: Capability CAPTURE\_FULL\_PAGE\_IR
- ►3.2.6: Capability CAPTURE\_FULL\_PAGE\_UV
- ►3.2.7: Capability READ\_ELECTRONIC
- ►3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS
- ►3.2.9: Capability COMPARE\_DOCUMENTS

### 4.2.2 Requirements for Application scenario - Self Service ►3.1.2

The required capabilities for this scenario are:

- ►3.2.1: Capability SCAN\_MRZ
- ►3.2.2: Capability SCAN\_CAN
- ►3.2.3: Capability SCAN\_DIGITAL\_SEAL
- ►3.2.4: Capability CAPTURE\_FULL\_PAGE\_VI
- ►3.2.5: Capability CAPTURE\_FULL\_PAGE\_IR
- ►3.2.6: Capability CAPTURE\_FULL\_PAGE\_UV
- ►3.2.7: Capability READ\_ELECTRONIC
- ►3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS

### 4.2.3 Requirements for Application scenario - Partially mobile ▶3.1.3

The required capabilities for this scenario are:

- ▶3.2.1: Capability SCAN\_MRZ
- ▶3.2.7: Capability READ\_ELECTRONIC
- ▶3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS
- ▶3.2.9: Capability COMPARE\_DOCUMENTS

### 4.2.4 Requirements for Application scenario - Fully mobile ▶3.1.4

The required capabilities for this scenario are:

- ▶3.2.1: Capability SCAN\_MRZ
- ▶3.2.2: Capability SCAN\_CAN
- ▶3.2.3: Capability SCAN\_DIGITAL\_SEAL
- ▶3.2.7: Capability READ\_ELECTRONIC
- ▶3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS
- ▶3.2.9: Capability COMPARE\_DOCUMENTS

### 4.2.5 Requirements for Application scenario - Kiosk ▶3.1.5

The required capabilities for this scenario are:

- ▶3.2.1: Capability SCAN\_MRZ
- ▶3.2.3: Capability SCAN\_DIGITAL\_SEAL
- ▶3.2.4: Capability CAPTURE\_FULL\_PAGE\_VI
- ▶3.2.5: Capability CAPTURE\_FULL\_PAGE\_IR
- ▶3.2.6: Capability CAPTURE\_FULL\_PAGE\_UV
- ▶3.2.7: Capability READ\_ELECTRONIC
- ▶3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS
- ▶3.2.9: Capability COMPARE\_DOCUMENTS

### 4.2.6 Requirements for Application scenario - Asylum registration ▶3.1.6

The required capabilities for this scenario are:

- ▶3.2.1: Capability SCAN\_MRZ
- ▶3.2.3: Capability SCAN\_DIGITAL\_SEAL
- ▶3.2.4: Capability CAPTURE\_FULL\_PAGE\_VI
- ▶3.2.5: Capability CAPTURE\_FULL\_PAGE\_IR
- ▶3.2.6: Capability CAPTURE\_FULL\_PAGE\_UV
- ▶3.2.7: Capability READ\_ELECTRONIC
- ▶3.2.8: Capability COMBINE\_DOCUMENT\_COMPONENTS
- ▶3.2.9: Capability COMPARE\_DOCUMENTS

## 4.3 Requirements for Capabilities

The following sections describe which capabilities necessitate the execution of which checks.

### 4.3.1 Requirements for Capability SCAN\_MRZ ▶3.2.1

An IS with this capability SHALL perform the following checks:

- ▶3.3.3: Check - Document Detection
- ▶3.3.4: Check - Document Model Identification
- ▶3.3.5: Check - MRZ consistency

### 4.3.2 Requirements for Capability SCAN\_CAN ▶3.2.2

An IS with this capability SHALL perform the following checks:

- ▶3.3.3: Check - Document Detection

### 4.3.3 Requirements for Capability SCAN\_DIGITAL\_SEAL ▶3.2.3

An IS with this capability SHALL perform the following checks:

- ▶3.3.3: Check - Document Detection
- ▶3.3.4: Check - Document Model Identification
- ▶3.3.18: Check - Digital Seal

### 4.3.4 Requirements for Capability CAPTURE\_FULL\_PAGE\_VI ▶3.2.4

An IS with this capability SHALL perform the following checks:

- All checks mandatory to ▶Section 4.3.1 and ▶Section 4.3.3.
- ▶3.3.7: Check - MRZ OCRB type face
- ▶3.3.8: Check - MRZ alignment
- ▶3.3.14: Check - Spectrally Selective Verification (VI)

### 4.3.5 Requirements for Capability CAPTURE\_FULL\_PAGE\_IR ▶3.2.5

An IS with this capability SHALL perform the following checks:

- All checks mandatory to ▶Section 4.3.1 .
- ▶3.3.7: Check - MRZ OCRB type face
- ▶3.3.8: Check - MRZ alignment
- ▶3.3.14: Check - Spectrally Selective Verification (IR)

This SHALL include at least one sub check of the type (**IR, AB, MR**). If applicable to the identified document model, this SHALL include at least one sub check of type (**IR, TR, ZZ**). See [BSI TR-03135-1],  Section 5.7.2.2: Spectrally selective checks for the interpretation of the terms in brackets.

### 4.3.6 Requirements for Capability CAPTURE\_FULL\_PAGE\_UV ▶3.2.6

An IS with this capability SHALL perform the following checks:

- ▶3.3.14: Check - Spectrally Selective Verification (UV )

This SHALL include at least one sub check of the type (**UV, BR, FU**). If applicable to the identified document model, this SHALL include at least one sub check of type (**UV, LU, ZZ**). See [BSI TR-03135-1],  Section 5.7.2.2: Spectrally selective checks for the interpretation of the terms in brackets.

### 4.3.7 Requirements for Capability READ\_ELECTRONIC ▶3.2.7

An IS with this capability SHALL perform the following checks:

- ▶3.3.15: Check - Electronic

### 4.3.8 Requirements for Capability COMBINE\_DOCUMENT\_COMPONENTS ▶3.2.8

An IS with this capability SHALL perform the following checks:

▶3.3.19: Check - Combined

### 4.3.9 Requirements for Capability COMPARE\_DOCUMENTS ▶3.2.9

An IS with this capability SHALL perform the following checks:

▶3.3.20: Check - Cross Document

## 4.4 Requirements for Checks

The following sections describe requirements for checks. If additional checks are supported, their execution is RECOMMENDED, but SHALL be decided by the operator. Any checks that are performed SHALL be represented by the respective elements in the logging schema according to [BSI TR-03135-1], Chapter 6: Logging scheme for operational monitoring.

### 4.4.1 Requirements for Check - Document ▶3.3.1

The requirements for this check are as follows:

- The false negative rate of the overall result of the document checking process per country SHOULD not exceed 2% for correctly placed, captured and identified documents.
- handling of defects as specified in [BSI TR-03135-1], Section 5.12: Handling and interpretation of defects

### 4.4.2 Requirements for Check - Optical ▶3.3.2

The requirements for this check are as follows:

The inspection system SHALL offer a configuration option that changes the `runMode` of the spectrally selective checks to `conditionallyIgnored` under certain conditions. To enable this feature, the inspection system SHALL support a whitelist of countries. This whitelist SHALL be configurable by the operator of the inspection system.

The change of the `runMode` of the spectrally selective checks SHALL only be applied if the following conditions are met:

- The document under inspection was issued by a country that is included in the whitelist.
- The ▶3.3.15: Check - Electronic was successful.

### 4.4.3 Requirements for Check - Spectrally Selective Verification ▶3.3.14

The requirements for this check are as follows:

Partial checks SHALL be performed according to the identified document model. The vendor and the operator SHALL come to an agreement on which partial checks SHALL use which `runMode`.

### 4.4.4 Requirements for Check - Issuer versus Nationality ▶3.3.9

The requirements for this check are as follows:

The IS SHALL be allowed to set the `runMode` of this check to `evaluatory`, unless the operator requires the `runMode` to be set to another value.

### 4.4.5 Requirements for Check - No Test Document ▶3.3.10

The requirements for this check are as follows:

The IS SHALL be allowed to set the `runMode` of this check to `evaluatory`, unless the operator requires the `runMode` to be set to another value.



#### 4.4.6 Requirements for Check - Primary Facial Image Embedding ▶3.3.11

The requirements for this check are as follows:

The IS SHALL be allowed to set the `runMode` of this check to `evaluatory`, unless the operator requires the `runMode` to be set to another value.

#### 4.4.7 Requirements for Check - MRZ versus Visual Zone ▶3.3.12

The requirements for this check are as follows:

The IS SHALL be allowed to set the `runMode` of this check to `evaluatory`, unless the operator requires the `runMode` to be set to another value.

#### 4.4.8 Requirements for Check - Visual Zone Consistency ▶3.3.13

The requirements for this check are as follows:

The IS SHALL be allowed to set the `runMode` of this check to `evaluatory`, unless the operator requires the `runMode` to be set to another value.

#### 4.4.9 Requirements for Check - MRZ consistency ▶3.3.5

The requirements for this check are as follows:

This check SHALL be omitted if no International Civil Aviation Organization (ICAO) compliant MRZ is expected for the identified document model.

#### 4.4.10 Requirements for Check - MRZ IR readability ▶3.3.6

The requirements for this check are as follows:

This check SHALL be omitted if no ICAO compliant MRZ is expected for the identified document model.

#### 4.4.11 Requirements for Check - MRZ OCRB type face ▶3.3.7

The requirements for this check are as follows:

This check SHALL be omitted if no ICAO compliant MRZ is expected for the identified document model.

#### 4.4.12 Requirements for Check - MRZ alignment ▶3.3.8

The requirements for this check are as follows:

This check SHALL be omitted if no ICAO compliant MRZ is expected for the identified document model.

#### 4.4.13 Requirements for Check - Electronic ▶3.3.15

The requirements for this check are as follows:

A recent Master list (see [BSI TR-03135-1] [↗](#) Section 4.2.1: Receiving Master Lists, Defect Lists, Document Signer Lists, Signer Lists and access authorization certificates) SHALL be used.

#### 4.4.14 Requirements for Check - Authenticity ▶3.3.16

The requirements for this check are as follows:

If one of the partial checks was performed with the result `successful`, the IS SHALL be allowed to skip the execution of the other partial checks, unless the operator requires the execution of all partial checks.

#### 4.4.15 Requirements for Check - CSCA Validity ▶3.3.17

The requirements for this check are as follows:

This check SHALL be executed with `runmode` set to `evaluatory`.

#### 4.4.16 Requirements for Check - Digital Seal ▶ 3.3.18

The requirements for this check are as follows:

A recent Signer List (see [BSI TR-03135-1]  Section 4.2.1: Receiving Master Lists, Defect Lists, Document Signer Lists, Signer Lists and access authorization certificates) for all domains (see [BSI TR-03129-2] ) relevant for the identified document model SHALL be used.

## 5 Conformity

In order to conform to this Technical Guideline, an Inspection System and Inspection Application SHALL completely implement and meet all requirements from ►Chapter 4 for the applicable application scenario.

# List of Abbreviations

| Abbreviation | Description                                 |
|--------------|---------------------------------------------|
| ABC          | Automated Border Control                    |
| CAN          | Card Access Number                          |
| eMRTD        | Electronic Machine Readable Travel Document |
| ICAO         | International Civil Aviation Organization   |
| IR           | infrared                                    |
| IS           | Inspection System                           |
| MRTD         | Machine Readable Travel Document            |
| MRZ          | Machine Readable Zone                       |
| RFID         | Radio-Frequency Identification              |
| UV           | ultraviolet                                 |
| VI           | white light                                 |
| XML          | Extensible Markup Language                  |

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